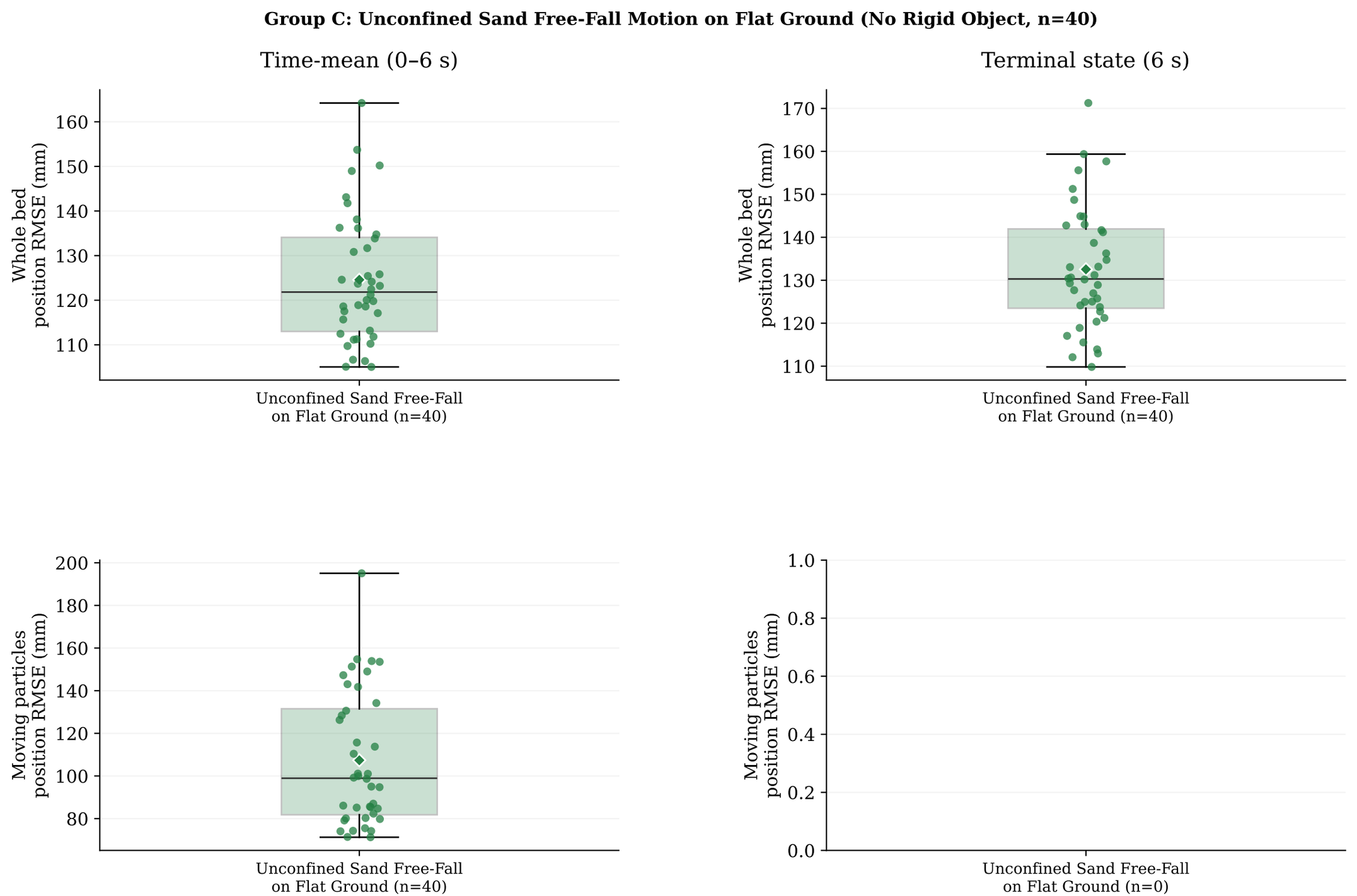

Figure 1c: Frozen object-model checkpoint 11,000. Unconfined 3D mid-air sand blocks/columns dropped onto open flat ground (no retaining box walls and no rigid object; n=40 held-out test trajectories). Top row: whole-bed position RMSE; bottom row: moving-particles position RMSE (>0.01 m/s).